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Protection via Separation

materials are based on security courses at other institutions

Separation, like Alcatraz?



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Resources That Need Protection



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- **Multiprogramming increased the need for protection**
- **Resources that need protection**
 - Memory
 - Shared IO devices
 - Shared programs and modules
 - Networks
 - Shared data

Methods of Separation



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- **Physical**
 - Multiple printers, disks, ...
- **Temporal**
 - Insecure before 12pm; secure after 12pm
- **Logical**
 - OS/environment provides isolation
- **Cryptographic**
 - Data and computation concealed cryptographically

Degrees of Separation



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- **None**
 - All resources are fully shared
- **Complete isolation**
 - Programs are completely oblivious to each other
- **Binary sharing**
 - Resources are either public or private
- **Limited sharing**
 - Fine-grained control
 - Authorization checked on a per-access basis
 - E.g., ACLs, capabilities
- **Usage control**
 - Control not just access to resources, but how they're used

Granularity of Separation



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- What are the objects that are being separated from each other?
- Programs
- Sets of programs
- Parts of a program
- OSes

Today's mechanism

- Hardware
 - Memory protection
- Software
 - Sandboxing
- Hardware + software
 - Virtual machines



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Memory protection

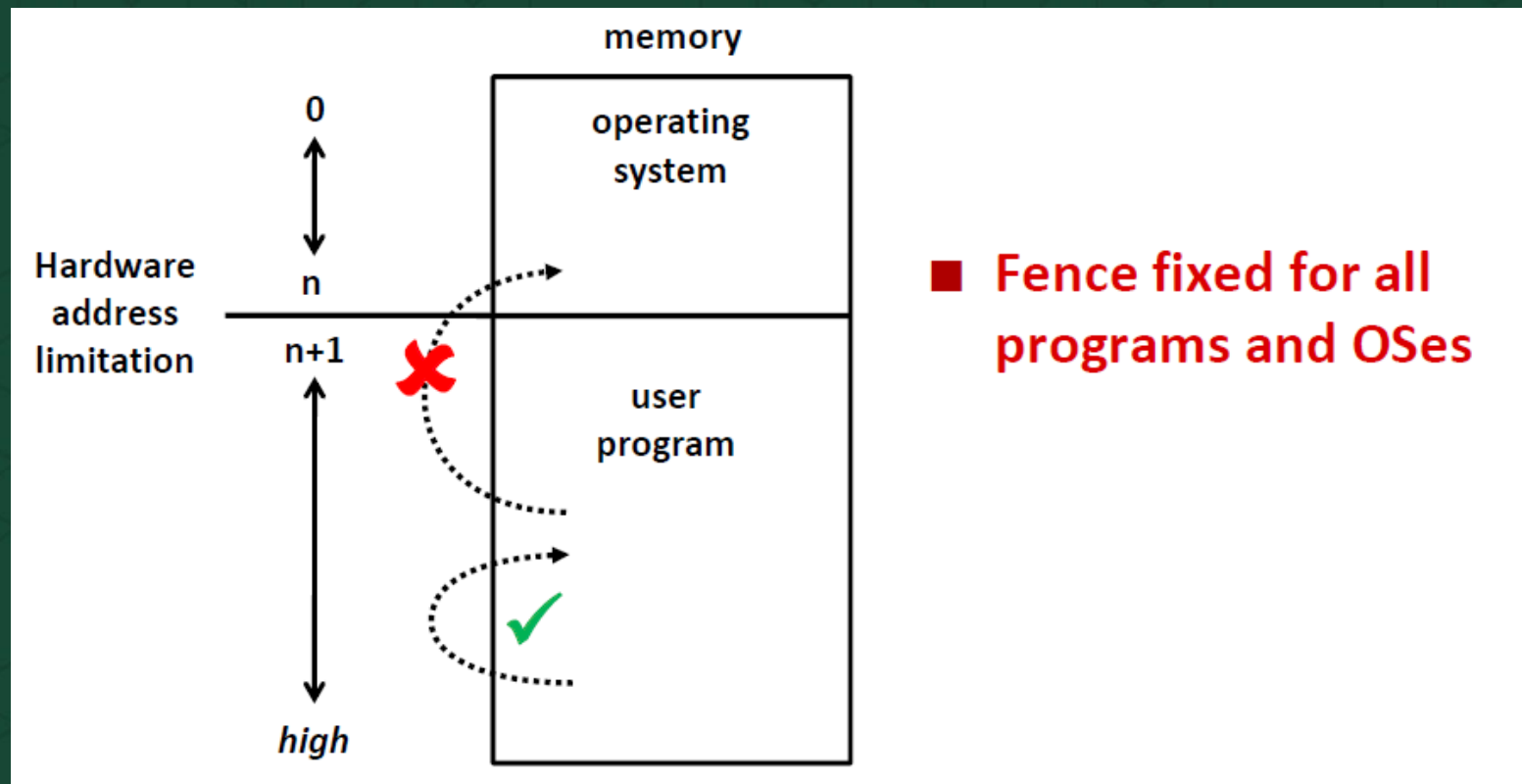
**Logical separation, between programs, using hardware
(and software)**

Memory Protection – Fence



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- **Boundary between user programs and operating system**

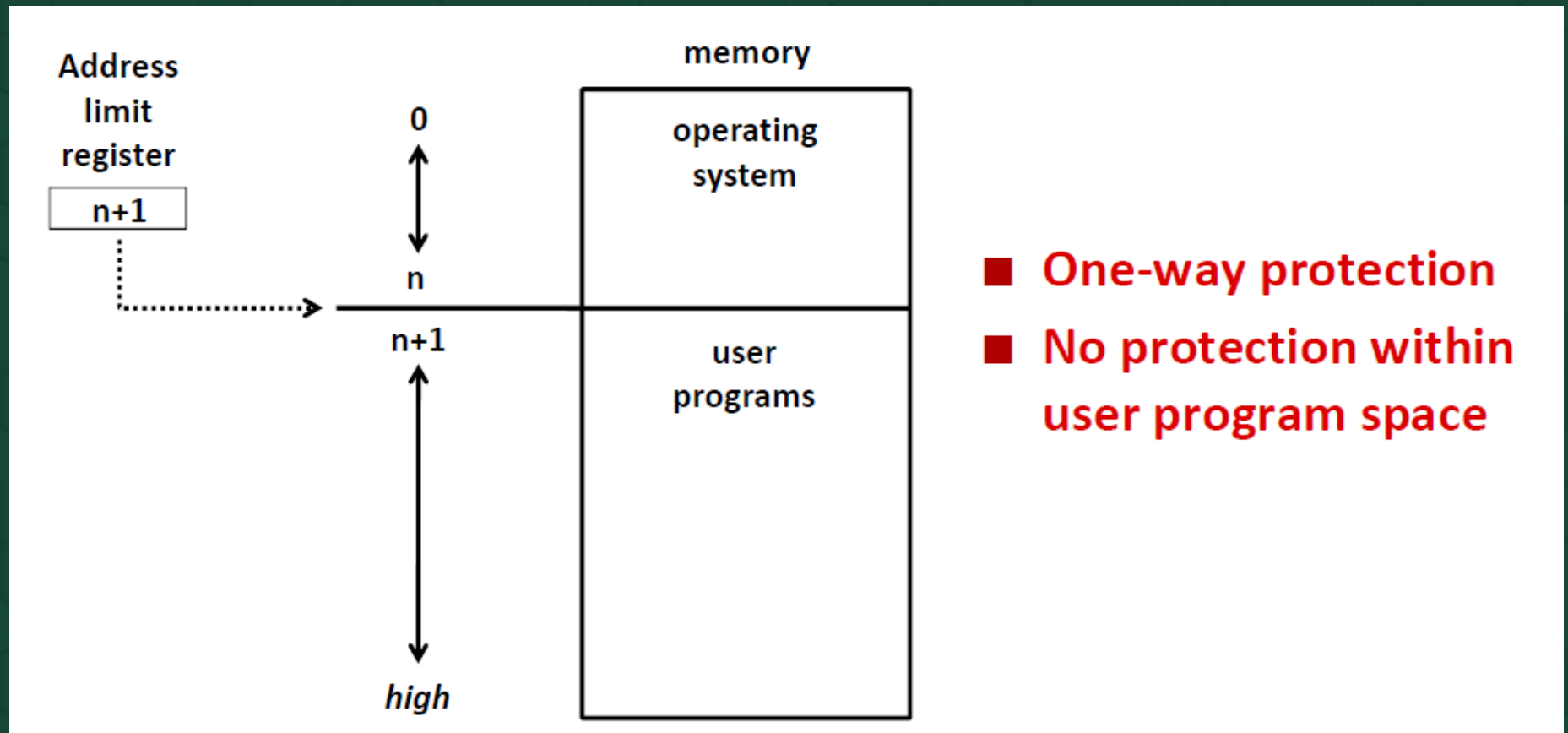


Memory Protection – Fence



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- A slight refinement

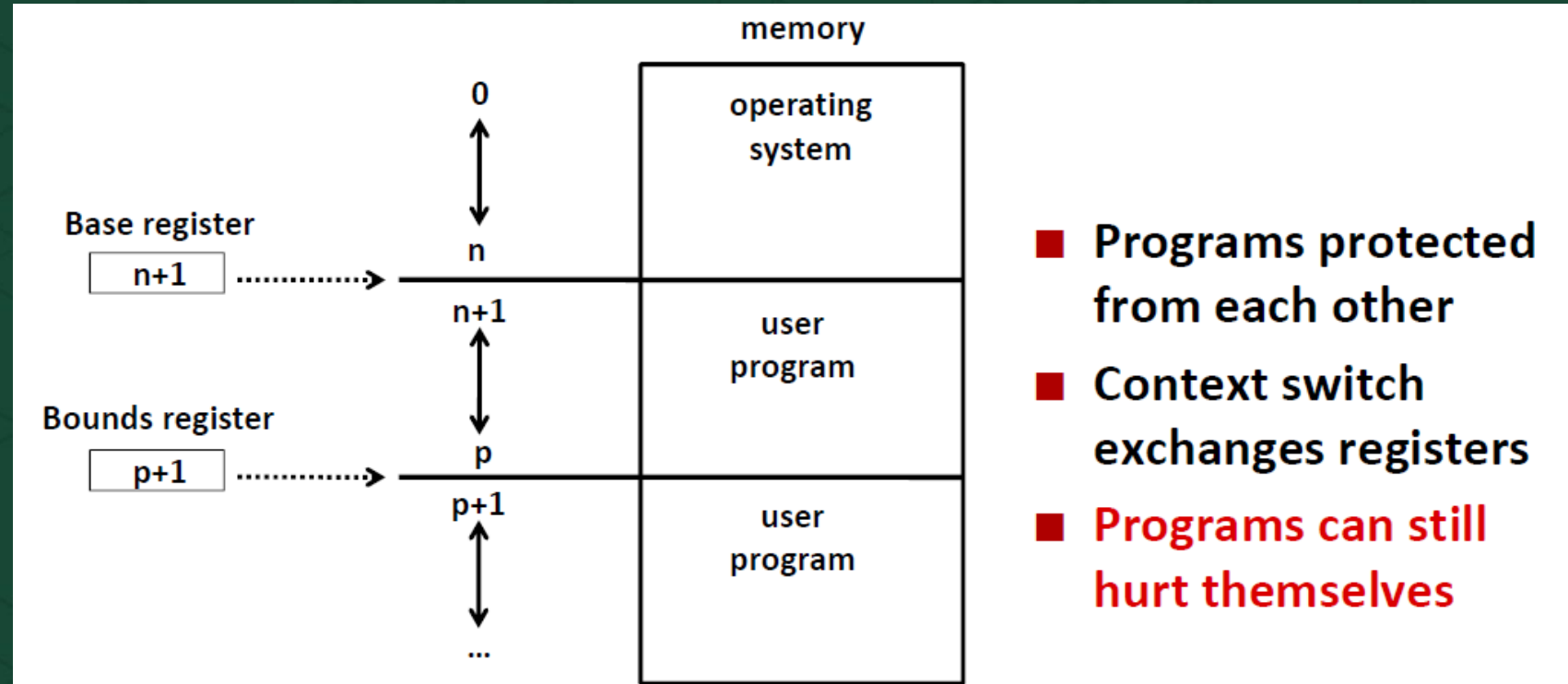


Memory Protection – Base/Bounds Registers



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- If one fence is good, two are better

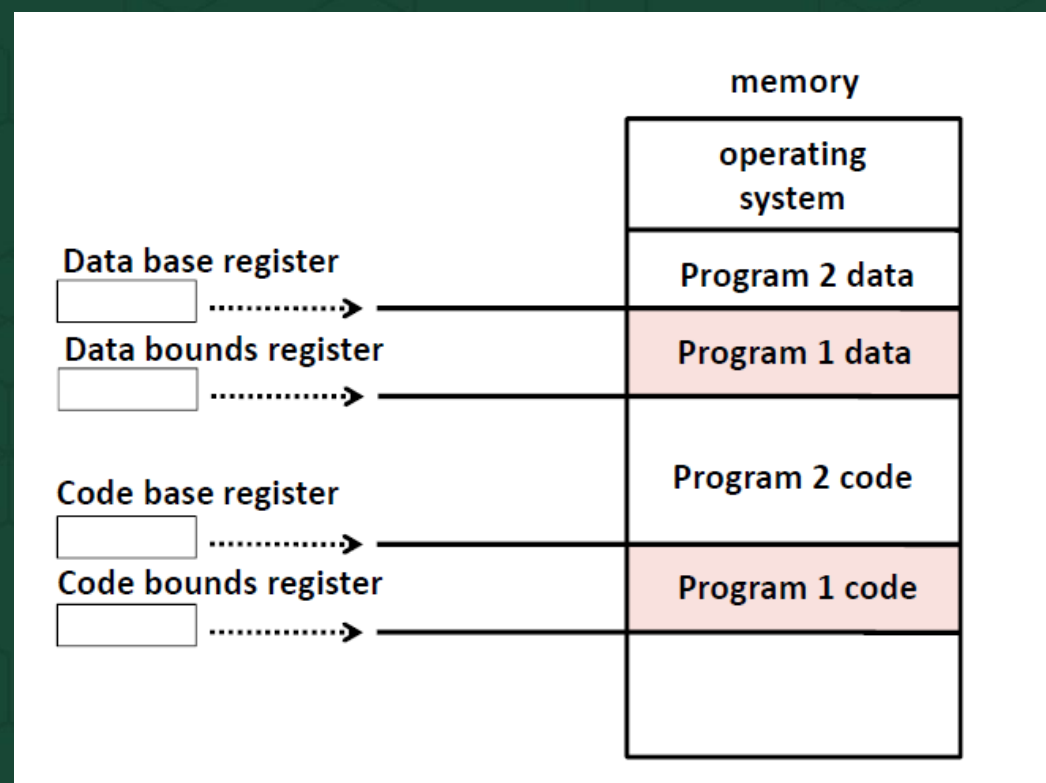


Memory Protection – Base/Bounds Registers



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- If two are good, four are better



Memory Protection – Base/Bounds Registers

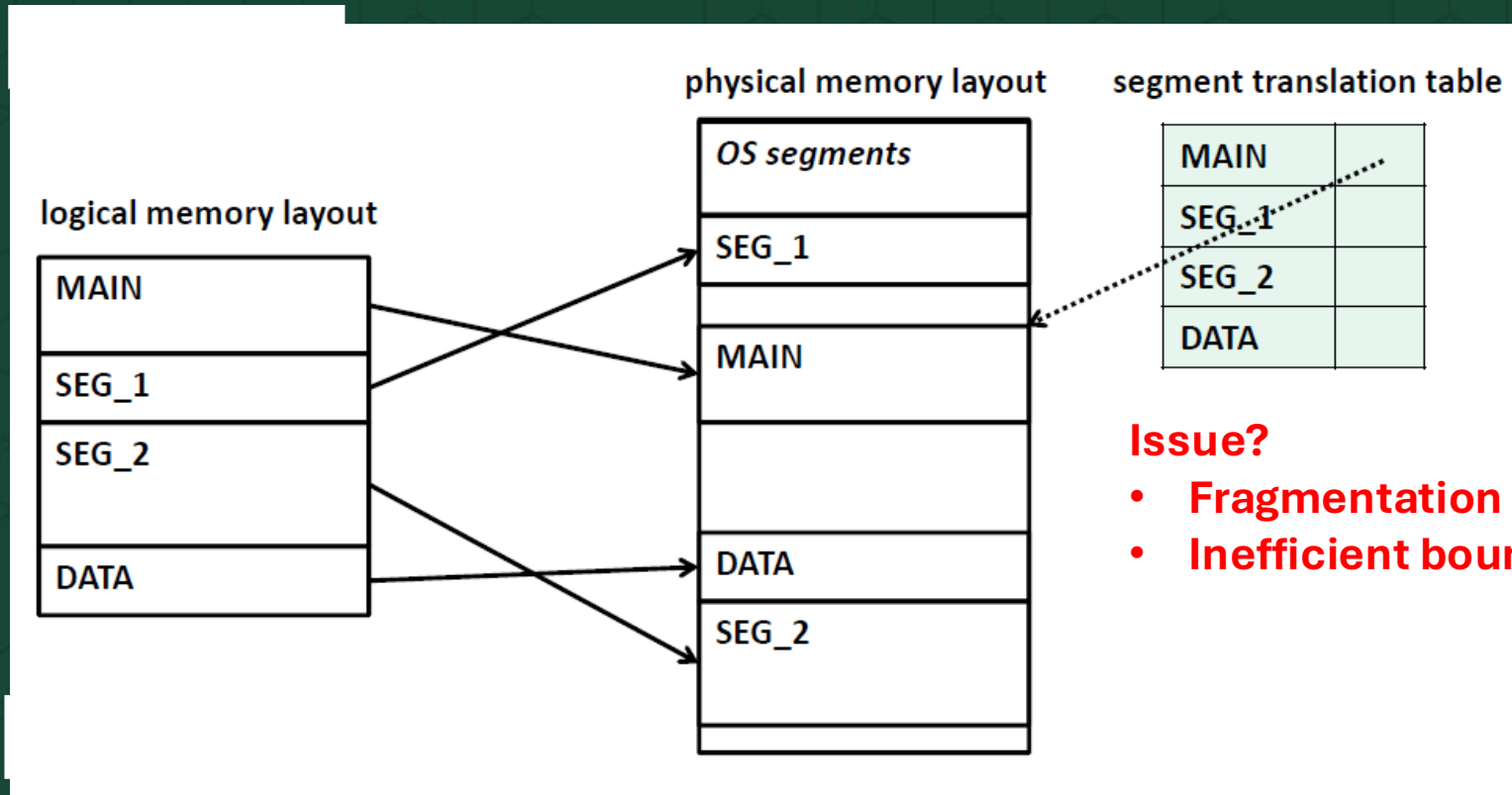


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- **If four are good, ... ?**
- E.g., separate data into read-only and read-write
 - Instructions have to include pointer to data space
 - In the limit, each word of memory can be in own domain, labeled as read-only, read-write, or execute

Memory Protection – Segmentation

- Program divided into many logical, named pieces
- Addresses have the form <name, offset>
- OS maintains translation table



Memory Protection – Paging



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- Program divided into equal-sized *pages*
- Addresses have the form $\langle page, offset \rangle$
- Is fragmentation a problem?
- Is boundary check a problem?
- Logical segmentation isn't preserved
 - Pages don't correspond to logical program segments

Discussion



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What is the difference between segmentation and paging?

Advantages? Drawbacks?

Can we combine them in modern systems?

How to combine them?

Memory Protection – Segmentation & Paging



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- Combine efficiency of paging with protection of segmentation
 - First divide into segments, then each segment into pages
- Intel x86 architecture implements paged segmentation ...
 - but not widely used
 - Windows/Linux today: paging, no segmentation



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Software Protection

**Software Fault Isolation $\approx$ memory protection in
software**

Software Fault Isolation (SFI) Goals



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- **Confine faults inside distrusted extensions**
 - codec shouldn't compromise media player
 - device driver shouldn't compromise kernel
 - plugin shouldn't compromise web browser
- **Allow for efficient cross-domain calls**
 - numerous calls between media player and codec
 - numerous calls between device driver and kernel

SFI: Main Idea



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- Allocate each untrusted program to a *fault domain* (contiguous memory space + unique identifier)
- Modify untrusted program to prevent it from writing or jumping to an address outside its fault domain

SFI: In More Detail



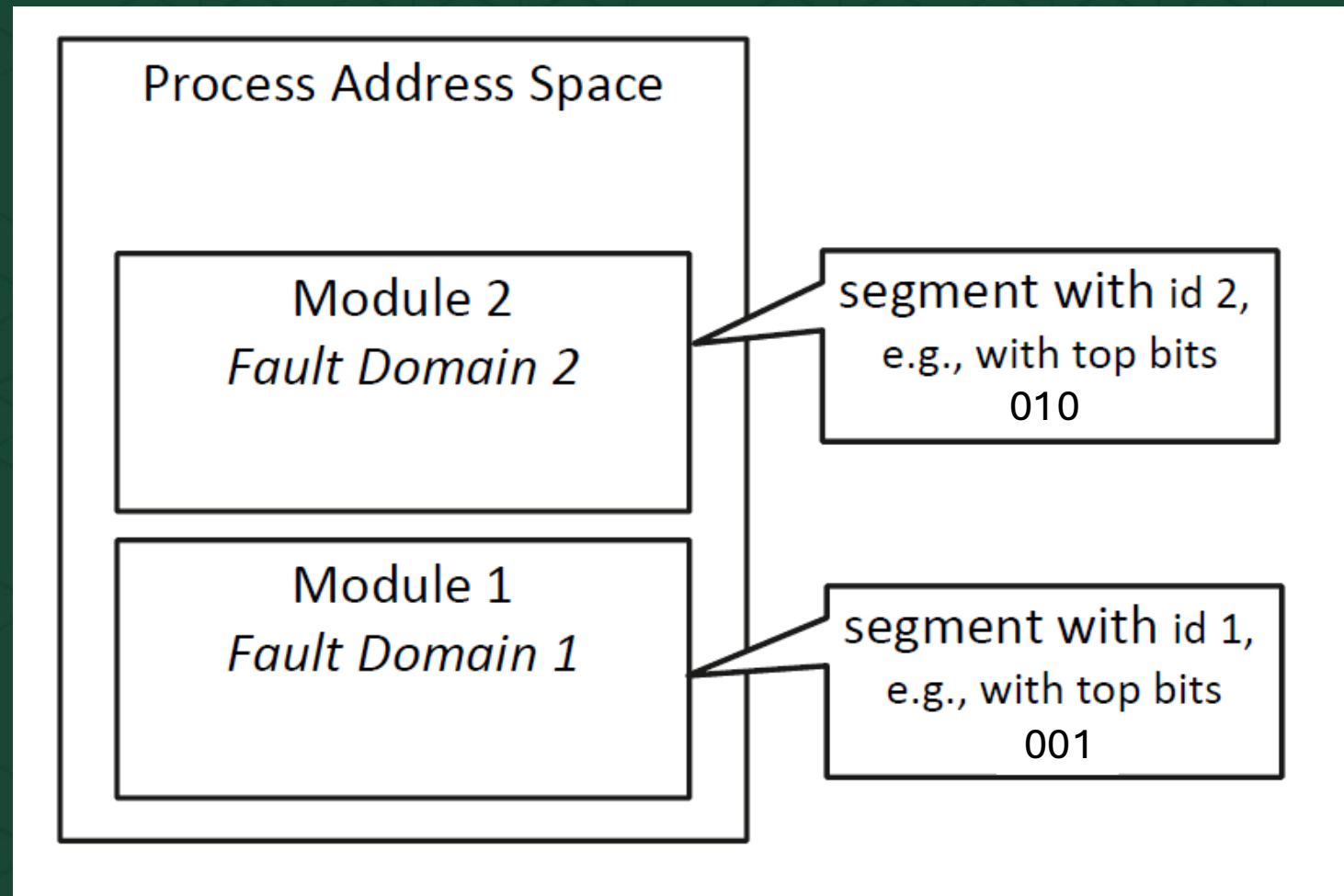
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- **Code divided into code and data segments**
 - Jump targets restricted to code segment
 - Data addresses restricted to data segment
- **Code and data segment addresses stored in dedicated registers**
- **Before executing any instruction that references memory**
 - Compare segment of memory reference to stored segment addresses
 - Stop execution if addresses aren't equal
- **Applications**
 - User-level file systems, Google's NativeClient

Main Idea



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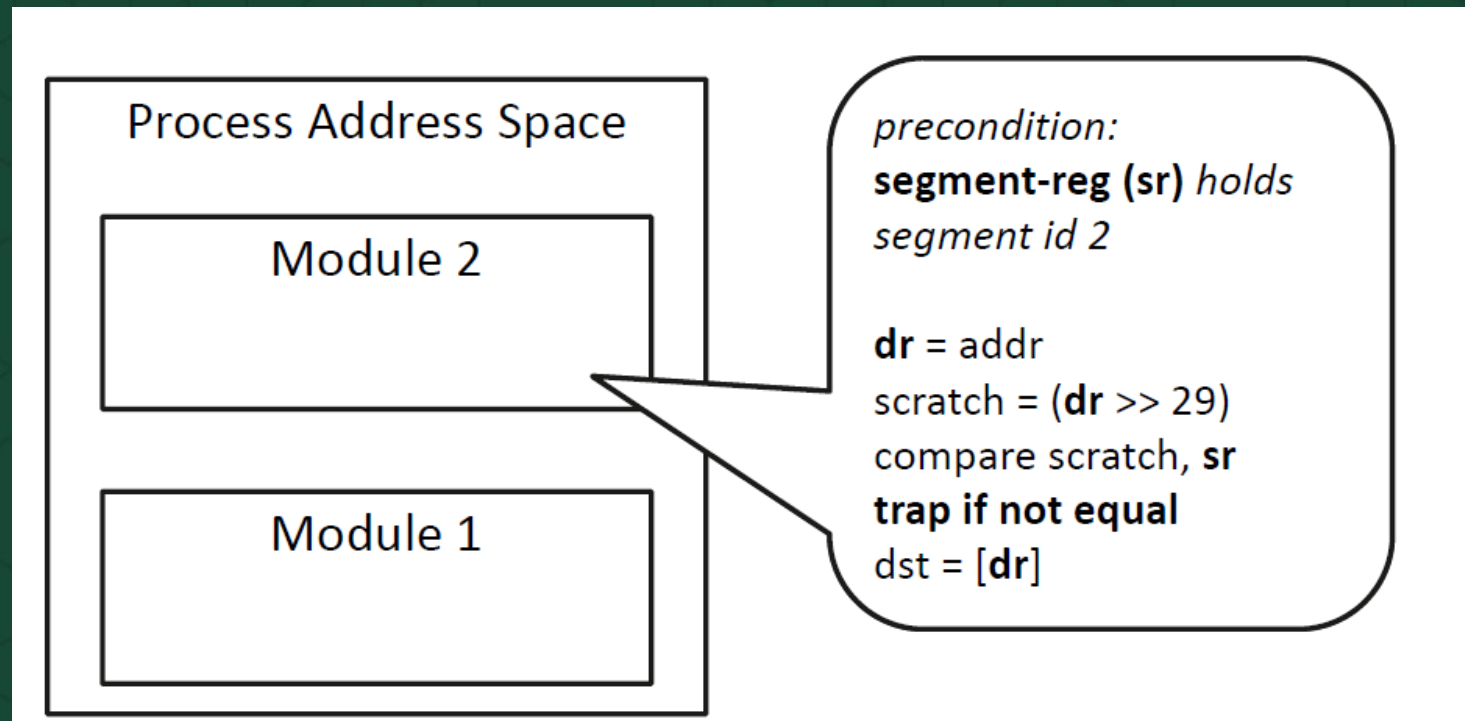


SFI Scheme: Segment Matching

- Check every mem access for matching seg ID
- Assume dedicated register **dr**, **sr**
 - not available to the program

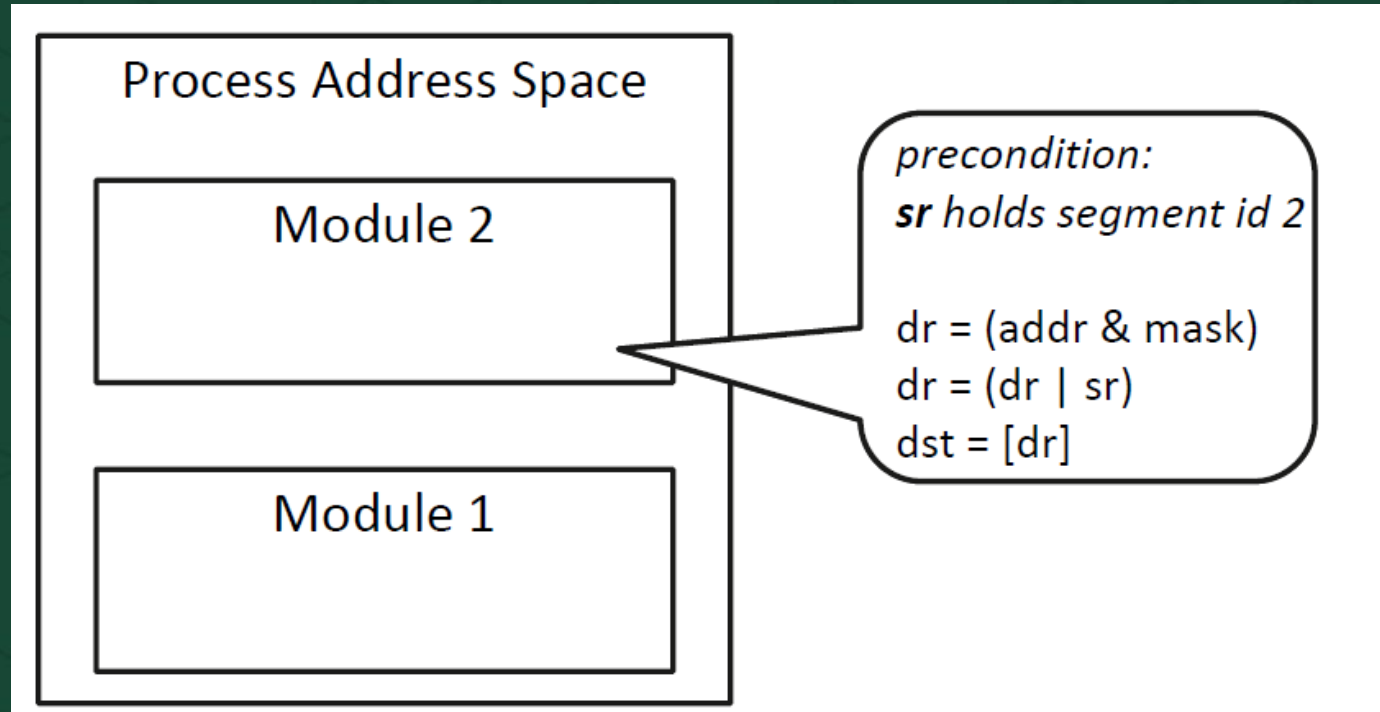


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SFI Scheme: Sandboxing

- Force top bits to match seg ID and continue
- No comparison is made



Segment Matching vs. Sandboxing



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Segment Matching

- **more instructions**
- **can pinpoint exact point of fault where segment ID doesn't match**

Sandboxing

- **fewer instructions**
- **just ensures memory access stays in region (crash is ok)**

Take Away: Reference Monitors



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- External reference monitor
 - e.g., OS component with hardware-support enforces memory separation among untrusted programs
- Inlined reference monitor
 - e.g., SFI mechanism enforces memory protection by rewriting untrusted program to ensure it accesses memory within its own space